

The Dance of the Chromosomes

I PUC · Ch 10 · Cell Cycle and Cell Division

Starring Bindu 👧 & Prof. BAIO 🦉

A BAIO storybook by Narendra Sir · Biology Lecturer



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One Cell, Many Cells

Prof. BAIO showed Bindu an onion root tip under the microscope. "Every big body starts as one cell. Cells grow and divide, again and again. This repeating story is the **cell cycle**." A human cell takes about 24 hours for one cycle, but **yeast** needs only 90 minutes.

> 💡 **REMEMBER**
>
> **Cell cycle: copy the genome, make cell parts, divide into two daughter cells.**



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The Busy Rest

"This cell looks quiet," said Bindu. "It is busiest of all!" laughed Prof. BAIO. In **interphase** the cell grows in **G1 phase**, copies its DNA in **S phase**, and makes proteins for division in **G2 phase**. Cells that stop dividing, like heart cells, rest in **G0**.

💡 REMEMBER

S phase: DNA doubles from 2C to 4C, but the chromosome number stays 2n.



Getting Ready

In **prophase** the tangled threads coiled into thick **chromosomes**. Each had two **sister chromatids** held together at a **centromere**. The centrosomes walked to opposite poles and threw out star-like **asters**. Slowly, the nuclear envelope disappeared.

💡 **REMEMBER**

Asters + spindle fibres = the mitotic apparatus.



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The Line-Up

In **metaphase**, spindle fibres caught each chromosome by its **kinetochore** and pulled it to the middle. All the chromosomes stood in one line, the **metaphase plate**. "Now is the best time to count them!" said Prof. BAIO.

💡 REMEMBER

Metaphase is the best stage to study the shape of chromosomes.



Tug of War

Snap! In **anaphase** every centromere split at once, and the sister chromatids raced to opposite poles, centromere first, arms trailing. In **telophase** they relaxed, and a new nuclear envelope formed around each group.

💡 REMEMBER

Telophase is the reverse of prophase.

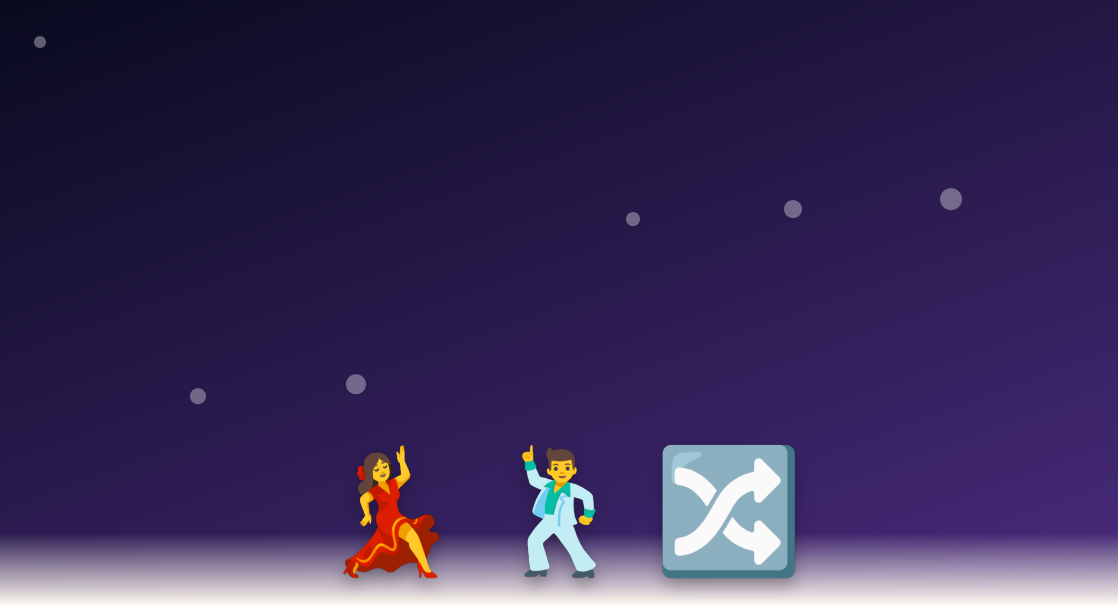


Two Ways to Split

Now the cytoplasm had to divide: **cytokinesis**. The animal cell pinched in with a **furrow**, like a belt tightening on a balloon. The plant cell, with its stiff wall, built a **cell plate** from the centre outwards. In coconut, nuclei divided without the cell dividing, making a **syncytium**.

💡 **REMEMBER**

Animal cell: furrow (outside in). Plant cell: cell plate (centre out) = middle lamella.



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The Special Dance

"To make gametes, cells do a special dance called **meiosis**," said Prof. BAIO. In prophase I, partner chromosomes paired up by **synapsis** and swapped pieces by **crossing over**, helped by the enzyme **recombinase**. X-shaped **chiasmata** showed where they had swapped.

💡 REMEMBER

Prophase I: Leptotene, Zygotene, Pachytene, Diplotene, Diakinesis.



Four New Cells

In meiosis I the partners separated, giving a **dyad** of two cells. Meiosis II worked like mitosis, and four haploid cells, a **tetrad**, were born. "At fertilisation two such cells join, so the chromosome number stays the same in every generation, and crossing over makes every child unique!"

💡 **REMEMBER**

Meiosis I is reductional; meiosis II is equational.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

cell cycle p.1 · yeast p.1 · interphase p.2 · G1 phase p.2 · S phase p.2 · G2 phase p.2 · G0 p.2 · prophase p.3 · chromosomes p.3 · sister chromatids p.3 · centromere p.3 · asters p.3 · metaphase p.4 · kinetochore p.4 · metaphase plate p.4 · anaphase p.5 · telophase p.5 · cytokinesis p.6 · furrow p.6 · cell plate p.6 · syncytium p.6 · meiosis p.7 · synapsis p.7 · crossing over p.7 · recombinase p.7 · chiasmata p.7 · dyad p.8 · tetrad p.8



QUICK QUIZ

Do you remember?

1. DNA is copied in:

- G1 phase
- S phase
- G2 phase
- M phase

2. The best stage to count chromosomes is:

- Prophase
- Metaphase
- Anaphase
- Telophase

3. Crossing over takes place in:

- Leptotene
- Zygotene
- Pachytene
- Diakinesis

4. Liquid endosperm of coconut is an example of:

- Syncytium
- Cell plate
- Tetrad
- Dyad

5. Homologous chromosomes separate in:

- Anaphase I
- Anaphase II
- Metaphase I
- Telophase II

Answers: 1-S phase · 2-Metaphase · 3-Pachytene · 4-Syncytium · 5-Anaphase I